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Geophysical Research Letters

RESEARCH LETTER
10.1029/2023GL107185

Effect of Nucleation Heterogeneity on Mineral Precipitation in Confined Environments

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Key Points:

- The proposed computational framework focuses on investigating the heterogeneity of nucleation and its effects on mineral precipitation
- The new model is validated by comparing with experimental data, demonstrating an agreement in both nucleation morphology and rate
- The new model integrates probabilistic nucleation, resulting in a notable enhancement in the accuracy of permeability predictions

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Yang, F., Guan, D., Starchenko, V., Yuan, K., Stack, A. G., & Ling, B. (2024). Effect of nucleation heterogeneity on mineral precipitation in confined environments. *Geophysical Research Letters*, *51*, e2023GL107185. https://doi.org/10.1029/2023GL107185

Received 18 NOV 2023
Accepted 15 APR 2024

Abstract The formation of new mineral phases in confined environments, especially in porous media, is crucial for various geological processes like mineralization and diagenesis. The nucleation and precipitation of minerals are initiated at the microscale through fluid-rock interaction, where dissolution of primary phases leads to supersaturated conditions and nucleation and growth of secondary ones. Previous research has focused primarily on either precipitation or nucleation, without fully exploring their combined impact. Our study introduces a computational framework that integrates classical nucleation theory with the micro-continuum method. We validated our model by comparing with experiments, and discovered that different surface nucleation rate changes the mode of precipitation from a preferential to uniform precipitate textures. Furthermore, our study uncovered that the conventional deterministic precipitation method tends to underestimate the permeability of the porous matrix. In contrast, the new framework significantly improves model accuracy by incorporating preferential precipitation and heterogeneous nucleation.

Plain Language Summary The formation of minerals in confined environments, especially in porous media, is important for both natural processes and industrial applications. The resulting solid phases can greatly affect the porosity and permeability of the porous structure. Many previous studies have focused on either nucleation or precipitation, we still have a limited understanding of how both processes interact at different scales. In this study, we integrated the Darcy-Brinkman-Stokes method and classical nucleation theory to investigate the interplay between nucleation, precipitation, species transport, and their impact on permeability. Our findings show that adjusting the surface nucleation rates can change the mode of mineral precipitation from preferential to uniform. We also found that the traditional deterministic precipitation method underestimates permeability in certain scenario. Therefore, considering probabilistic nucleation is crucial when there is preferential precipitation and heterogeneous nucleation on the interfaces between the minerals and surrounding liquid under reactive flow conditions.

1. Introduction

Rock-fluid interactions, which involve mineral nucleation and precipitation, are ubiquitously observed in natural and engineering systems. In a practical setting, the efficient management of these phenomena could have significant benefits for applications such as carbon dioxide mineralization (Hills et al., 2020; Snæbjörnsdóttir et al., 2020), oil and gas extraction (Crabtree et al., 1999; Cui et al., 2021; Esteves et al., 2023), as well as remediation of environmental contamination (Jiang et al., 2019; Sharma & Kumar, 2020). Recent studies have shown that considerable amounts of mineral precipitates (*e.g.*, barite and Fe(III) (hydro)oxides) were formed during the hydraulic fracturing process, which may potentially alter the permeability and transport pathways within the shale matrix, thereby impacting the long-term hydrocarbon production (Hakala et al., 2021; Weber et al., 2021; Xiong et al., 2021, 2022). As such, comprehending the mechanisms of mineral nucleation and precipitation under reactive flow conditions in porous media is critical to unveiling geological diagenetic processing and optimizing industrial applications.

Mineral nucleation and precipitation in porous media represent a complex phenomenon that encompasses various physics processes, including fluid flow, transport of species, interfacial processes, and complex geochemical reactions (Stack, 2015; Steefel & Maher, 2009). Modern imaging techniques such as computerized tomography (CT), magnetic resonance imaging (MRI), and X-ray nano-tomography (XnT) have been developed to monitor and analyze intricate rock-fluid phenomenon. Prior research has identified two distinct categories of mineral

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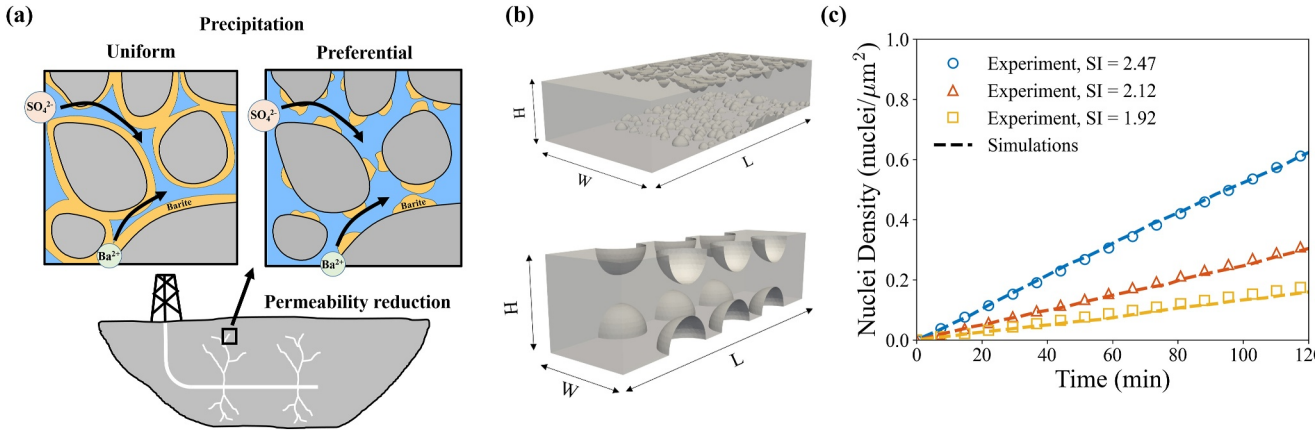


Figure 1. (a) Uniform and preferential precipitation. (b) Schematics of simulation systems (narrow pore and porous matrix). (c) Comparison of nuclei density of barite crystals on the channel wall as a function of time between DBS simulations and XnT experiments (SI = 1.92, 2.12, 2.47, respectively).

precipitation modes, preferential precipitation, and uniform precipitation, encountered during reactive flow in porous media (Borgia et al., 2012; Emmanuel et al., 2010; Stack, 2014, 2015). In the preferential precipitation mode, the solid phase tends to form scattered crystals on the rock-fluid interface, forming a heterogeneous precipitation texture, whereas the uniform precipitation mode results in a uniform coating of precipitate (Figure 1a). However, conducting in-situ measurements of nucleation and precipitation of minerals as well as transport of species in porous media remains a challenging and expensive task for experimental studies. As a result, numerical simulation methods, such as the Lattice Boltzmann method (LBM) (Chen et al., 2014; Kang et al., 2010), smoothed particle hydrodynamics (SPH) (Tartakovsky, Meakin, Scheibe, & Eichler West, 2007; Tartakovsky, Meakin, Scheibe, & Wood, 2007), and computational fluid dynamics (CFD) (Molins et al., 2021; Soulaine et al., 2018; Starchenko et al., 2016), have been developed as powerful tools for comprehending the transport of species and the evolution of various phases in pores and porous media. Among different numerical methods, one of the promising numerical simulation approaches is a micro-continuum simulation framework based on the Darcy-Brinkman-Stokes (DBS) equation, which possesses several advantages, including flexible geometry for different phases, implicit tracking of interfacial reactions, and multiscale capability (Maes et al., 2022; Soulaine et al., 2019; Steefel et al., 2015; Yang et al., 2021).

In macro-scale modeling of mineral precipitation processes, the mineral nucleation process has frequently been oversimplified. It has been commonly assumed that precipitation occurs automatically when the solution is oversaturated or a pre-existing solid phase is present in the system (Borgia et al., 2012; Bringedal et al., 2020; Chen et al., 2013; Nogues et al., 2013; Steefel & Hu, 2022; Tartakovsky, Meakin, Scheibe, & Wood, 2007). Consequently, deterministic precipitation methods have been extensively used to examine mineral precipitation in porous media. These methods generally consider precipitation as a consistent thickening of the solid phases that covers the rock-fluid interfaces (Deng et al., 2016; Emmanuel & Berkowitz, 2005; Xiong et al., 2021). However, recent research has shown that the incorporation of nucleation with mineral precipitation is of great interest and has significant impacts on evolution of porous structures (Noiriel et al., 2021; Nooraiepour et al., 2021; Weber et al., 2021). For instance, Fazeli et al. explored the effect of probabilistic mineral nucleation processes using the Lattice Boltzmann Method (LBM) coupled with an induction-time-based nucleation model (Fazeli et al., 2020). Their findings revealed that the probabilistic process of mineral nucleation had a considerable impact on the porosity-permeability relationship in reactive flow.

This manuscript commences by providing a detailed description of the coupling method between the mineral nucleation process and the reactive flow, followed by a validation of the solver's effectiveness through X-ray nano-tomography (XnT) experiments. Subsequently, we examine the behaviors and mechanisms of mineral precipitation by modeling reactive flow in a narrow pore. Additionally, we discover that the mineral precipitation mode in a porous matrix may be influenced by the heterogeneous nucleation rate, resulting in a transition from preferential precipitation to uniform precipitation. The results highlight the critical role played by the nucleation-governed morphology of precipitated solid phases in determining the permeability of porous media, as a combined effect of fluid flow and species transport. Thus, the findings underscore how the microscopic nucleation of

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mineral crystals influences mesoscopic precipitation in pores and ultimately results in macroscopic permeability differences in porous media.

2. Materials and Methods

2.1. Nucleation and Preferential Precipitation

In this study, we aim at studying precipitation processed occurring on heterogeneous surfaces which is frequently observed in real geological formations with complex mineral components. Precipitation dynamically changes the solid fraction in the modeling domain, and the uniform precipitation process can be quantified by:

$$\frac{\partial m_s}{\partial t} = M_s R_u, \tag{1}$$

where m_s [g] is the solid mass, t [s] is time, M_s [g/mol] is the molar mass for the solid phase and R_u [mol/s] is the uniform growth rate within certain computational cell. For precipitation case, $R_u > 0$ and m_s increases as the reaction progresses. The uniform growth rate R_u is a function of reaction constant, solid-fluid interface density and local concentration:

$$R_u = k_i a_v^i (c - c_{sat})\, V_c, \tag{2}$$

where k_i [m/s] is the precipitation reaction rate constant for the mineral-i precipitates on certain type of solid surface, a_v^i [m^2/m^3] is the mineral-i volumetric solid-fluid interface density of for the corresponding reaction, V_c [m^3] is the volume of computational cell, c [mol/m^3] is the concentration of species and c_{sat} [mol/m^3] is the saturation concentration at the interface. When the local concentration is higher than the saturation concentration, that is, $c > c_{sat}$, the total solid precipitation rate $M_s R_u > 0$ and m_s increases as time increases. This approach of modeling uniform precipitation has been widely employed in the previous studies by assuming all liquid-solid interfaces are available for precipitation at the start of simulations. To ensure the sharpness of the interface, we used a criterion method from (Yang et al., 2021) to limit the distance of the diffuse interface to one layer of computational cell.

In this study, to model the preferential precipitation at pore scale, we revise Equation 1 by introducing nucleation:

$$\frac{\partial m_s}{\partial t} = M_s R \quad \text{and} \quad R = \begin{cases} R_n, & \text{before nucleation} \\ (R_u + R_n), & \text{after nucleation} \end{cases} \tag{3}$$

where R_n [mol/s] is the effective nucleation rate, it is associated with the local averaged probability (ϕ_{NCL}) of mineral nucleation within a unit volume for a given nucleation period t_{NCL}:

$$R_n = M_{nucleus} \cdot \frac{\phi_{NCL}}{t_{NCL}}, \tag{4}$$

where $M_{nucleus}$ [mol/nucleus] is the mole number of per nucleus, ϕ_{NCL} [nuclei] is the amount of formed nuclei on a substrate in a computational volume cell during t_{NCL}. The nucleation rate can be computed following the Classic Nucleation Theory (CNT):

$$\frac{\phi_{NCL}}{t_{NCL}} = A_{c,s} \cdot J, \tag{5}$$

$$J = A \cdot exp\left(\frac{-\Delta G^*}{k_b T}\right), \tag{6}$$

$$\Delta G^* = \frac{16\pi\gamma^3 V_{mol}^2}{3(2.303\mathrm{SI} \times k_b T)^2}, \tag{7}$$

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where J [nuclei/(m^2·s)] is the nucleation rate, $A_{c,s}$ [m^2] is the surface area of the substrate, A [nuclei/(m^2·s)] is a pre-factor accounting for collision frequency of solution ions with each other and the substrate, γ [mJ/μm^2] is the surface free energy, V_{mol} [m^3/mol] is the molar volume of the solid phase, SI is the saturation index of the solution. For barite, SI = $log(a_{Ba}a_{SO4}/K_{sp})$, where $K_{sp} = a_{Ba}a_{SO4}$ at equilibrium. k_b [mJ/K] is the Boltzmann constant, T [K] is the temperature of the solution. Here, we utilized a probabilistic method similar to previous work (Yang et al., 2022) at each unit time ($t_{NCL} = 1$ s) to identify which computational cell had nucleation occurred within it.

According to Equation 3, the change in local solid mass is influenced by both precipitation and nucleation, and such treatment enables the solver to model the preferential precipitation. In this study, we assumed that the precipitation rates of the solvent on the substrate or boundaries are negligible in the absence of nucleation, which is supported by experimental findings (De Yoreo & Vekilov, 2003; Nielsen et al., 2014; Noiriel et al., 2021; Yuan et al., 2021). Hence, during the initial stage of rock-fluid interaction, the precipitation is practically absent until nuclei form on the substrate/boundary in the over-saturated solution based on the probability described by Equations 5–7. These nuclei create new solid-fluid interfaces within a very small volume, resulting in a local volumetric solid-fluid interface density, a_v, for precipitation to occur. As a result, the growth reaction rate is significantly enhanced at these preferential precipitation locations.

2.2. Micro-Continuum Scale Numerical Framework

In order to simulate the transport of species, fluid flow as well as precipitation process, we used the DBS approach developed in our previous study similar to the mpFoam solver (Yang et al., 2021). Here, we briefly list the main governing equations for fluid flow and transport of species. The comprehensive details of the implementation of governing equations and numerical schemes can be found in previous work (Yang et al., 2021). The micro-continuum framework quantifies fluid and solid as volumetric ratio within a micro-scale control volume, namely:

$$\varepsilon_f = \frac{V_f}{V_\Omega} \quad \text{and} \quad \varepsilon_s = \frac{V_s}{V_\Omega}, \tag{8}$$

where ε_f and ε_s are the volumetric fractions of the fluid and solid phases within the control volume, respectively, V_f and V_s are the volume [m^3] of the fluid and solid, and V_Ω [m^3] is the total volume of the micro-scale control volume. The precipitation of this single reactive species is described by Equation 3, and using the micro-continuum representation:

$$\frac{\partial \varepsilon_s \rho_s}{\partial t} = \begin{cases} M_s R_n / V_\Omega, & \text{before nucleation} \\ M_s (R_u + R_n)/V_\Omega, & \text{after nucleation} \end{cases}. \tag{9}$$

The methods mentioned above were implemented using an open source CFD package OpenFOAM with additional implementation of the nucleation model. All simulations were performed with Lawrence Berkley National Laboratory's NERSC Cori and Perlmutter clusters with a minimum of 64 CPU cores and 256 GB memories for each case.

2.3. Simulation Systems

In this study, we examine the dynamics of reactive transport in both a narrow pore and a porous matrix, considering the occurrence of crystal nucleation and solid phase precipitation. Two simulation models are constructed to understand how the reactive flow and transport of species is affected by the nucleation and precipitation processes.

The first model simulates the clogging process of a narrow pore by the precipitation of minerals. The dimensions of the simulation system ($L \times W \times H$) in this study are set as a 100 $\times$ 50 $\times$ 20 μm. Figure 1b shows the schematics of simulation system. To mimic the natural porous media, the second model is constructed by packed quarter-spheres that contain pores and throats (Figure 1b), and the dynamic of the pore-throat morphology during reaction is simulated. The flow channel dimensions were set as a 120 $\times$ 30 $\times$ 30 μm. The spherical obstacle has a radius of 10 μm and are located away from each other with a distance of 30 μm in x-direction. The top, bottom and sides walls of the simulation system are set to symmetry boundary conditions to replicate a sphere-packed porous matrix. The nucleation of minerals was enabled on the quarter spherical objects' surfaces. For both models

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described above, the porosity and permeability of the porous system could be estimated using the following forms,

$$\varepsilon_{f,V} = 1 - \varepsilon_{s,V} = \int_V \varepsilon_f(x)\, dv / V_{sys}, \tag{10}$$

$$K = -\frac{\mu_f \dot{Q}}{A_{ch} \Delta \overline{p}}, \tag{11}$$

where $\varepsilon_{f,V}$ is the porosity of the whole simulation domain, K [m^2] is the permeability, $\dot{Q}$ [m^3/s] is the volumetric flow rate, A_{ch} [m^2] is the flow channel cross section area.

A set of parameters mimicking the real minerals such as barite is used, which is within the range of a group of minerals observed in precipitation processes (Weber et al., 2021; Yang et al., 2022; Yuan et al., 2021). In current study, the general input parameters for crystal nucleation and growth are the molecular diffusion coefficient $D = 1 \times 10^{-9}$ m^2/s, solid phase molar mass $M_s = 200$ g/mol, solution concentration SI = 2.1 with solubility product constant $pK_{sp} = -log_{10}\, K_{sp} = 10$, $c_{sat} = 0.01$ mol/m^3, surface energy $\gamma = 33$ mJ/m^2, molar volume $V_{mol} = 5 \times 10^{-5}$ m^3/mol. The initial solid volume fraction for an initial nucleate within an individual control volume was set as $\varepsilon_{s,\mathrm{n}} = 1 \times 10^{-3}$, which is small enough comparing to the overall size of the system. Similarly, the $M_{nucleus}$ could be given by combining this value with the molar volume and control volume size, $\frac{\varepsilon_{s,\mathrm{n}} V_\Omega}{V_{mol}}$. The saturation index is sufficiently high such that heterogeneous nucleation and growth of barite is observed in the above references, but not homogeneous nucleation.

2.4. Experimental Validation

In this study, we validate the capability of the *mpFoam* with fast X-ray nano-tomography (XnT) experiments, which were conducted at National Synchrotron Light Source II (NSLS II) located in the Brookhaven National Laboratory, NY, USA. The flow cell mainly consisted of a solution mixer, syringe pump, and a quartz capillary tube. During the XnT experiments, the over-saturated barium sulfate solution was pre-mixed in the mixer and pumped into the capillary tube for the crystallization process. The capillary tube had an inner diameter of 0.1 mm and the flow rate was set as 100 μL/min. More details of the XnT experiments and their results could be found in previous work (Yuan et al., 2021).

The simulation system was a 100 μm long and 100 μm diameter cylinder injected with barium sulfate solution of concentration $c_b = 0.0933$, 0.117, 0.176 mol/m^3 (0.0933, 0.117, 0.176 mM), which corresponds to supersaturation SI = 1.92, 2.12, 2.47 for $K_{sp} = 10^{-9.98}$ (barite) (Blount, 1977). The mainstream computational cell size was set to be 1 μm for the validation case. The parameters for nucleation used in Equation 3: molar volume $V_{mol} = 5.211 \times 10^{-5}$ m^3/mol, $lnA = 20.19$ nuclei/m^2/s, $\gamma = 32.67$ mJ/m^2 and room temperature $T = 298$ K (Yuan et al., 2021). The lnA and γ are the fitting parameters from XnT experimental measurements given by the CNT. The flow profile was derived from the well-known Hagen-Poiseuille flow solution, which was implemented in the following form:

$$\overline{\boldsymbol{u}}(z) = \frac{3}{2}\overline{\boldsymbol{u}}_0 \left[1 - \left(\frac{R_p - z}{R_p}\right)^2\right] \tag{12}$$

where R_p is the radius of the quartz capillary tube, $\overline{\boldsymbol{u}}_0$ is the uniform flow velocity based on the flow rate, z is the distance from the cylinder wall.

Figure 1c shows the comparison between the nuclei density of barite crystal formed on tube wall in simulation and experimental measurements from XnT for different SI. The nuclei density was measured by counting the total amount of crystals formed on a fixed area of the wall and divided the observed surface area at different time. The results of the simulation (both nucleation rate and total nuclei amount) matched within the standard deviation (<3%) from the experimental results. The comparison of evolution of solid phase on tube wall between simulations and experiments (Figure S1 and S2 in Supporting Information S1) as well as more details on the validation

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case could be found in the Supporting Information. Additional validation regarding the nucleation rate prediction the *mpFoam* solver was presented in (Yang et al., 2022).

3. Results and Discussion

3.1. Precipitation Within Narrow Pore

To understand how the nucleation and precipitation of minerals affect the fluid flow and transport of species in narrow pore, we first fixed the nucleation rate and transport parameters in the system and tuned the precipitation reaction rate at the liquid-solid interface. To characterize the relationship between precipitation and transport of species, the effect of dimensionless Damköhler number ($Da_I = \frac{kH^2}{\overline{u_0}}$ and $Da_{II} = \frac{kH}{D}$) was considered. In this study, we mainly focused on the effect of Da_{II} on the mineral nucleation and precipitation. Hence, in the rest of the paper, we use Da to represent Da_{II} for convenience. The characteristic length scale H of the simulation system is 20 μm (height of the simulation domain) and the diffusion coefficient of solid species in solution is fixed. Hence, the Damköhler number of the simulation system was tuned by changing the k for mineral precipitation reaction (Equation 2). During the simulation, both the solid phase volume fraction $\overline{\phi}_{s,p} = V_{solid}/V_{sys}$ and liquid phase volume fraction of the simulation system were monitored. The inlet pressure of the system was set as 0.05 Pa to maintain a moderate initial flow rate, while the outlet pressure was set to the ambient pressure. In this way, the fluid flow was driven by the pressure difference between the inlet and outlet.

The simulation results indicate that once the solid phases nucleate on the walls, their growth is primarily regulated by the precipitation reaction. As precipitation directly influences the morphology of the solid phase, it gradually modifies the structure of the narrow pore's flow channel, consequently impacting the fluid flow and species transport through the pore. Figures 2a and 2b display the contour plot depicting the evolutionary timeline of the precipitated solid phases, starting from the initial nucleation stage, for various Da numbers. These plots were obtained by capturing data along the simulation system's central plain on the y-direction. It was observed that the precipitated solid phases in the lower Da number case (Da = 0.25) were distributed more evenly on the pore walls compared to the higher Da number case (Da = 2.5). In the scenario with a lower Da number, the precipitation thickness remains relatively constant from the upstream to the downstream. On the other hand, in the case with a higher Da number, a narrow pore throat forms near the inlet, with the solid phases primarily precipitating near the front of the pore and only a few nucleation sites at the rear half of the pore.

Such distinct behaviors of mineral precipitation are likely to be caused by the competition between the transport of species and precipitation processes. For the high Da number case, the initial nucleus formed near the inlet had a higher growth rate due to the fast transport of species from the inlet. However, as they grew, a pore throat was formed and the pathway for fluid flow and transport phenomena was partially blocked (see Figures 2c and 2d), which hindered the growth of the downstream mineral solid phase. Another observation from the simulations was the uneven mineral nucleation site distribution. In the higher Da number case, majority of the solute in the solution was consumed by the early precipitation of mineral crystals closer to the inlet, which greatly reduced the concentration of the solution downstream. Consequently, according to Equations 6 and 7, the nucleation rate (a function of concentration, Equation 6) on the downstream wall surface decreased and led to fewer nuclei forming (see Figure 2f). In contrast, as for the lower Da number case, the decrease of nucleation rate on the downstream wall surface was only moderate as shown in Figure 2e.

As mentioned above, the nucleation and growth of minerals could affect the fluid flow and transport of species through the narrow pore. To quantify the effect of different precipitation modes on the transport efficiency of the pore, we examined the evolution of the permeability of pore structures as a function of solid volume fraction (reverse of porosity), as shown in Figure 2g. At the initial stage, the permeabilities of the pore in different Da number cases showed similar decline trend. However, as the pore throat was narrowed in the higher Da number case, the permeability drastically decreased compared to the lower Da number case. This deviation of permeabilities signals that, at this stage, the fluid flow and transport of species are mainly controlled by the morphology of the precipitated solid phase within the narrow pore. Note that, when solid phase volume fraction was the same (0.2) for both cases, the permeability of the pore in the Da = 0.25 case is eight times higher than in the Da = 2.5 case, which is a significant variation in permeability for the same porosity. To further evaluate this deviation of behaviors, the pore throat size (located at the narrowest cross-section of the pore) was plotted in Figure 2h. The pore throat was around 11 μm for the Da = 0.25 case, and 5.48 μm for the Da = 2.5 case at $\varepsilon_{s,V} = 0.2$. The

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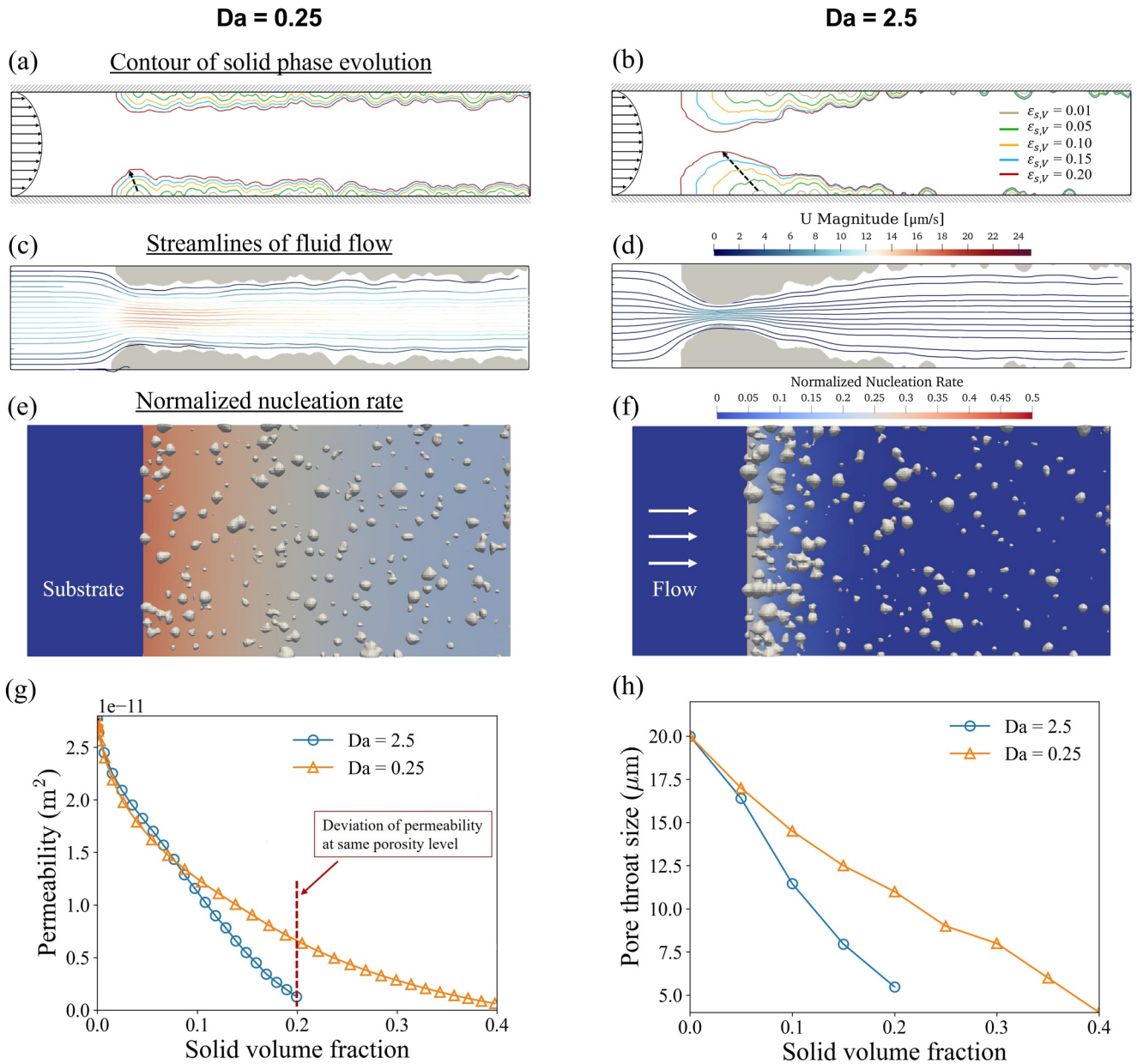


Figure 2. (a) and (b) Contour plot of the evolution history of precipitated solid phase for Da = 0.25 and 2.5. (c) and (d) Streamline plot of fluid flow. (e) and (f) Normalized nucleation rate on the pore wall for solid volume fraction at 0.02. (g) Permeability of the narrow pore as a function of the precipitated solid volume fraction for different Da numbers. (h) Minimum pore throat size as a function of precipitated solid volume fraction.

difference in the pore throat size causes a pressure drop of fluid flow passing through the throat. Consequently, the permeability of the pore varies with Da number due to the different morphology of the precipitated solid phase.

Normally, the nucleation of minerals would show heterogeneity on different rock-fluid interfaces. For example, the nucleation rate, crystal shape, size, and orientation may vary on different types of rock surfaces. In order to understand how the heterogeneity of nucleation behaviors may affect the morphology of the precipitated solid, we explored the effect of nucleation rate on the mineral precipitation process in the narrow pore. By tuning the pre-factor A of Equation 6, the nucleation rate of the pore wall surface was altered accordingly. The Da number in this set of simulations was fixed as 1.0 for considering both the effects of transport and reaction. One intriguing discovery from these simulations was that the pore with a higher nucleation rate exhibited a higher permeability for fluid flow and more efficient transport of species, as illustrated in Figure 3a. This observation appears to

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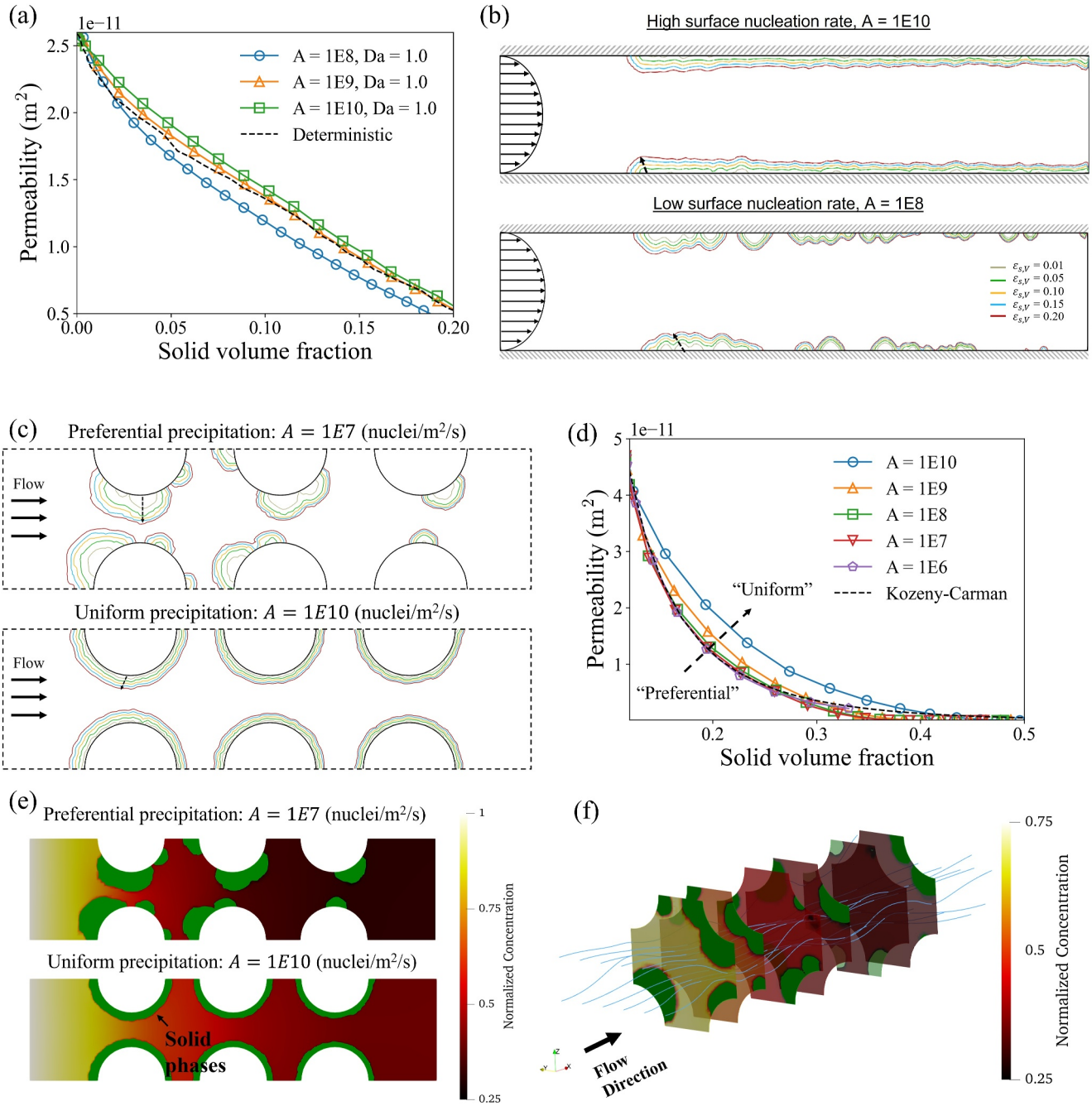


Figure 3. (a) Permeability of the narrow pore as a function of the precipitated solid volume fraction for different initial nucleation rates. (b) Contour plots of the evolution history of precipitated solid phase within narrow pores for different initial nucleation rates. (c) Contour plots of the evolution history of precipitated solid phase in porous matrix for different initial nucleation rates. (d) Permeability of the porous matrix as a function of the precipitated solid volume fraction for different initial nucleation rates. (e) Concentration distribution in porous matrix for $A = 1E7$ and $1E10$ (nuclei/m^2/s) cases. Precipitated solid phases was plotted as green color. (f) 3D heatmap plot of the fluid flow streamlines (blue lines) and normalized concentration distribution after "preferential precipitation" occurred ($A = 1E7$, solid: green color) within porous matrix. The fluid flow pathway was significantly disturbed by the dispersed growth of mineral crystals.

contradict the intuition that an increased number of nuclei would lead to a larger volume of precipitated solid phase, resulting in a reduction in the transport efficiency of the pore. Upon further examination of the results, we discovered that this observation is due to the differences in the total reactive surface area provided by the initial nucleation site and the preferential growing direction of the precipitated solid phase. When the nucleation rate is higher, the solution tends to form more nucleation sites, providing additional initial sites for the growth reaction.

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Consequently, the resulting solid phases are more evenly distributed within the pore, as shown in Figure 3b. In contrast, the lower nucleation rate case still had a limited number of crystals available for interfacial precipitation. As a result, these crystals grew at a faster rate due to the abundant solvent in the solution, leading to a rougher surface and further reduction in the pore throat size. Thus, the pore with a lower nucleation rate experienced a faster decline in permeability, in contrast to intuition. This observation highlights the crucial role of probabilistic nucleation in the precipitation of minerals within pore networks, as previously discussed by Fazeli and Deng (Deng et al., 2021, 2022; Fazeli et al., 2020). It emphasizes the importance of considering nucleation effects in studies involving interactions between rock-liquid phases, especially in the presence of precipitation. In addition, it was found that the pore permeability predicted by the deterministic precipitation method (Figure 3a, see Supporting Information S1 for more details) is closer to the high nucleation rate case, which also obtained a higher permeability than the pore with lower nucleation rate.

3.2. Precipitation Within a Porous Matrix

To understand how the nucleation of minerals interacts with a porous matrix, we investigated the evolution of the precipitated solid phase in the structure consisting of spherical arrays. The basic input parameters were similar to the previous sections. To mimic the heterogeneity of the rock surfaces, the nucleation rate on the solid surface was studied by tuning the prefactor A value. The study range of prefactor A was set from $O(10^6)$ to $O(10^{10})$, which represented both ultraslow and ultra-fast mineral nucleation behaviors in a porous matrix. The Da number in this set of simulations was fixed as 1.0 as well.

Remarkably, we observe a transition of the growth mode from preferential precipitation to uniform precipitation by solely tuning the surface nucleation rate. Figure 3c illustrates the solid phase evolution contours for $A = 1 \times 10^7$ nuclei/m^2/s. It is observed that the precipitated mineral phases in this lower A value case formed dispersed crystals/grains within the porous matrix. Similar to the above, as the A value increased, more porous structure surfaces were enabled for precipitation. Consequently, the precipitated mineral phases formed a uniform coating instead of dispersed crystals/grains. This shift in precipitation mode within the pore also resulted in different behaviors of how the permeability of the porous matrix evolved. Figure 3d plots the permeability of a porous matrix with varying initial nucleation rates as a function of the solid volume fraction. The widely used Kozeny-Carman (KC) correlation was adopted in the previous work (Steefel et al., 2015), which could be presented in the following form,

$$K = K_0 \frac{\left(1 - \varepsilon_{f,0}\right)^2}{\left(1 - \varepsilon_{f,V}\right)^2} \left(\frac{\varepsilon_{f,V}}{\varepsilon_{f,0}}\right)^3, \tag{13}$$

where $\varepsilon_{f,0}$ and K_0 is the porosity and permeability of porous matrix before precipitation occurs. It is observed that, as the initial nucleation rate increased, the decrease of permeability during the precipitation of minerals in the porous matrix was slowed and deviated from the Kozeny-Carman correlation results. Such results were mainly due to the dispersed distribution of mineral crystals/grains, which limited the ability of precipitated solid phases to cover the porous structure surface. Instead, the precipitated solid phases grew toward the space occupied by the liquid phases (see Figures 3e and 3f, Figure S5 in Supporting Information S1). The change in the porous matrix led to a smaller pore throat size and increased tortuosity. As a result, the preferential precipitation mode had a greater impact on reducing permeability compared to the uniform precipitation mode. Interestingly, the Kozeny-Carman correlation was found to be more accurate in evaluating the permeability of the preferential precipitation mode, while underestimating the permeability of the uniform precipitation mode. These findings shed light on the factors influencing nucleation and precipitation modes in real porous media. However, other factors like surface defects, surface charges, and multi-species precipitation/dissolution may also play a role and should be explored in future studies beyond the scope of this research.

4. Conclusion

This study presents a numerical investigation of the mechanism of mineral nucleation and precipitation, as well as reactive flow within confined environments (pore and porous matrix). To achieve this, a Darcy-Brinkman-Stokes model was utilized, in conjunction with classic nucleation theory. The model was initially validated against the results of X-ray nano-tomography experiments. Sequentially, to comprehend how mineral nucleation and

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precipitation behaviors impact fluid flow and species transport in porous media, two porous structure models were established and analyzed while varying a set of parameters. The findings suggest that the traditional deterministic precipitation modeling method may overestimate the permeability of porous media if the rock surface has a low nucleation rate for certain mineral crystals. Overall, the model has demonstrated its efficacy in simulating multi-physics systems, encompassing mineral nucleation, reactive flow, species transport, and precipitation, amongst other phenomena in confined environments.

The findings of this study underscore the importance of accounting for intrinsic heterogeneous nucleation and its impact on rock-fluid interactions, particularly in instances of occurrence of reactive flow and precipitation in porous structure. However, further research is necessary to fully comprehend how additional factors, including complex geometry and multiple surface reactions, may influence phase change processes and transport phenomena within porous media. The outcomes of this study are anticipated to provide insights into the mechanism of mineral nucleation and precipitation in porous media, thereby aiding in the design of future experiments.

Data Availability Statement

The numerical solver and simulation cases used in this study are open-sourced and distributed through a public Zenodo repository at Yang et al. (2024).

Acknowledgments
This work was supported by the International Collaboration Program of Chinese Academy of Sciences (025GJHZ2023016MI, F. Yang), the Strategic Priority Research Program of Chinese Academy of Sciences (XDB0620102), and the National Natural Science Foundation of China (42272158, B. Ling). The preliminary data and the research concept of this work were partially supported by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences, Chemical Sciences, Geosciences, and Biosciences Division. This research used resources of the Compute and Data Environment for Science (CADES) at the Oak Ridge National Laboratory, which is supported by the Office of Science of the U.S. Department of Energy under Contract No. DE-AC05-00OR22725, as well as the National Energy Research Scientific Computing Center (NERSC) clusters, a U. S. Department of Energy Office of Science User Facility operated under Contract No. DE-AC02-05CH11231.

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